

Formal Record of Dissertation Meeting

Student's Responsibility

It is the student's responsibility to take notes of the meeting and forward them to the supervisor in a timely manner.

Student:	Mykhailo Shpyl	Supervisor:	Dr. Vini Vijayan
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Programme:	MSc Artificial Intelligence and Big Data Analytics
Date:	4 June 2026 (Meeting 1)

Key Points Discussed (must include timetable for research and training record):
Project direction confirmed. Discussed the proposed explainable AI pipeline for academic integrity verification (AI text detection plus argument-aware verification question generation). Supervisor was positive on the topic, the tool choices, and the literature, and confirmed it is a research-oriented project where the value is in investigating the approach and reporting findings rather than guaranteeing a fully production-ready result. Dataset approach. Plan agreed: use an existing human-written essay dataset (BAWE) as the base, then generate AI-written versions using a local open-source model. To avoid the model learning irrelevant signals, AI generations will be matched on topic, keywords, and target length so the detector learns writing style rather than length or topic differences. Classification simplified to two classes. Supervisor raised the concern that “partially AI” and “fully AI” (and human vs partially AI) overlap heavily and are hard for NLP to separate reliably, partly because regular AI users carry an AI-like style into their own writing. Agreed to reduce the classification to two classes (human vs AI) rather than three or five. NLP tools / models. Discussed the detection model (a DeBERTa / RoBERTa transformer combined with stylometric features such as next-word-prediction-based perplexity). Supervisor will re-read the proposal and review the model details (BERT / RoBERTa / DeBERTa differences). Compute resources. Local generation last year took around four months on a laptop GPU (RTX 4060). Supervisor will ask the coordinator (Aoife) about access to ATU HPC / cloud (Hyperforms cloud) to speed up dataset generation and training. Literature review guidance. Restrict to recent papers only, roughly 2021–2026 (ideally 2023–2026) given the pace of the field. When using any AI tool to find papers, do not copy the reference or link directly; take the title, find the real paper, read it, and only then cite it. Writing strategy. Because the timeline is only ~13 weeks and the document carries high marks, start writing early. Begin the Introduction chapter after the next meeting, then the Literature Review (the most time-consuming chapter). Chapters can be revised later. Assessment format. Possible change from viva to recorded presentation this year; supervisor will confirm with the coordinator. Not a concern at this stage. Meeting cadence. Weekly supervision meetings, online, fixed for Tuesdays at 12:00. Flexible to reschedule within the same week if needed. Meeting invite to be sent by supervisor. Indicative timetable: Wk 1–2 data + environment; Wk 3–4 detection model; Wk 5–6 explainability + argument mining; Wk 7–8 question generation (two backends) + Bloom's classifier; Wk 9–10 integration + evaluation; Wk 11–13 writing and submission. Writing of Introduction and Literature Review to run in parallel from after Meeting 2.

Action items for Student:

- Prepare a system architecture (diagram) and a short presentation of the tools, the dataset, and the expected output of the system, for discussion at the next meeting.
- Begin creating the dataset now: download BAWE and start generating matched AI-written versions (topic, keywords, and length controlled).
- Set up the development environment so implementation can begin.
- Start drafting the Introduction chapter after the next meeting; begin the Literature Review using only recent (2021–2026) papers with manually verified references.
- Check the dissertation template on the Blackboard dissertation module against the template shared in the OneDrive folder; if different, upload the correct one.
- Complete and upload this Formal Record of Meeting to the OneDrive folder (week 1).

Action items for Supervisor:

- Re-read the proposal and review the detection model details (BERT / RoBERTa / DeBERTa) and revert with any questions or suggestions.
- Ask the coordinator (Aoife) about access to ATU HPC / cloud computing resources for dataset generation and model training.
- Confirm the assessment format for this cohort (viva versus recorded presentation).
- Send the recurring meeting invitation for Tuesdays at 12:00.

Date of next meeting:

Tuesday 9 June 2026, 12:00 (online)

Issues requiring immediate attention:

Compute capacity. Dataset generation on a local laptop GPU is slow (took ~4 months previously). Access to ATU HPC / cloud would materially de-risk the timeline and should be confirmed early (supervisor following up with coordinator).

Confirm scope change in writing. Reduction from multi-class to two-class (human vs AI) classification should be reflected in the architecture and the Introduction chapter.